Project: Worked examples
Title: Material strength margin
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the factors of safety of a material against an applied equivalent stress, on yield and, when given, on ultimate strength. It does not check fatigue or stability.

WHAT IT CALCULATES	RELATION
Factor of safety against yield and against ultimate	$n_y = f_y/\sigma$; $n_u = f_u/\sigma$
Governing factor, the ultimate margin scaled to the yield basis by $k_u \geq 1$	$n_{gov} = \min(n_y, n_u/k_u)$
Utilisation against the required factor	$u = n_{req}/n_{gov}$

Basis: Design factor and factor of safety[1](§1-10, 11th ed.).

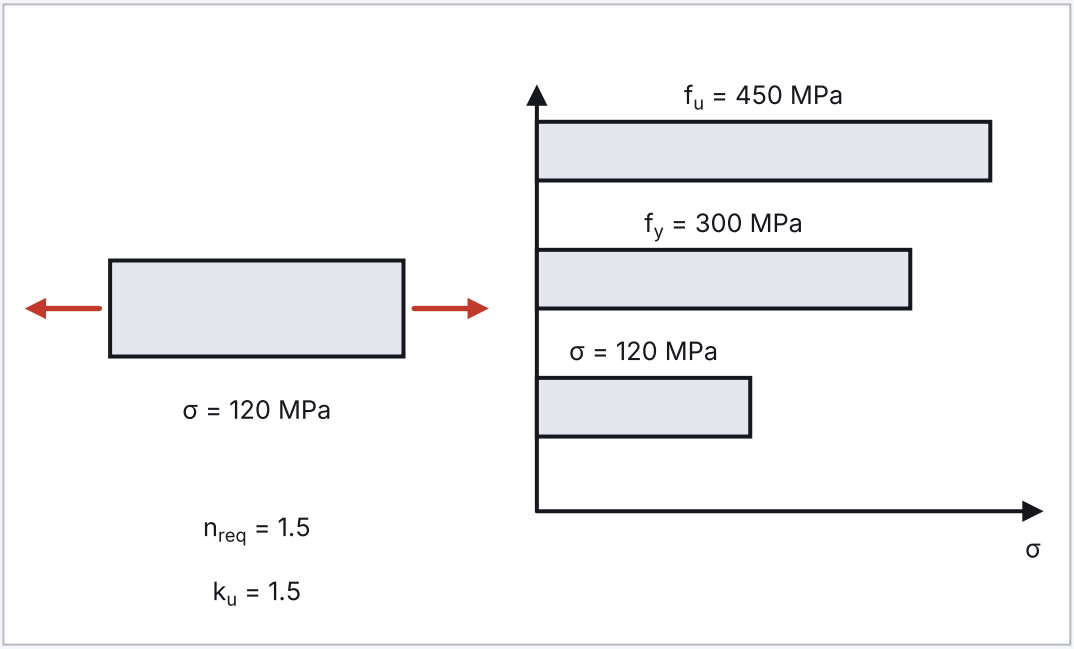


Figure 1 — A piece of the material under the applied stress, and the stresses as bars (drawn representatively, not compared). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
σ	Applied (equivalent) stress	120	MPa	spec: LOAD-SCHEDULE-01	0.01 … 5000 MPa
f_y	Yield strength	300	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa
f_u	Ultimate tensile strength (optional)	450	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_u	Ratio of the ultimate to the yield safety factor required	1.5		spec: SPEC-MECH-03 design criteria	1 … 3
n_{req}	Required factor of safety on yield	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. The load is static and the applied stress is the equivalent stress at the critical point. (Fatigue and stability are separate checks.)

4. Working

[5] $n_y = f_y / \sigma$ [1](§1-10)

= 300 MPa / 120 MPa

= 2.5

Factor of safety on yield

[7] $n_u = f_u / \sigma$

= 450 MPa / 120 MPa

= 3.75

Factor of safety on ultimate strength

[8] $n_{gov} = \min(n_y, n_u / k_u)$

= min(2.5, 3.75 / 1.5)

= 2.5

Governing factor (ultimate margin scaled to the yield basis)

[9] $u = n_{req} / n_{gov}$

= 1.5 / 2.5

= 0.6

Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
n_{gov}	2.5	Governing factor
n_y	2.5	Factor on yield
n_u	3.75	Factor on ultimate
u	0.60	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Required factor of safety against the governing factor	1.5	2.5	0.60	[1](§1-10)

7. Applicability

None.

8. Verification

Template mech.material-margin@1.1.0 Content hash 0b1c4346dd0d669e Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.